Daily Drilling Report
SPD19B-03
4/13/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
19.0	16"	40.00	72.83	31.70	106.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	221,095.38	7,075.48	4,453,869.97		16.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,441.00	1,478.00		37.00	11.82	3.1

Operations at Report Time
RIH directional BHA#6 @ 700m.

Operations Summary
Directional drilling 16" hole section f/ 1441m to 1478m. Due to Low ROP, POOH to surface. Observe Motor Sleeve stabilizer back off and was Loose. Try to change sleeve stabilizer, no success. Make decision to change motor.

Operations next 24hrs
RIH directional BHA#6 to bottom. Resume drilling to TD. Sweep hole.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/29/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	4/12/2014	6/2/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	3/22/2014	5/22/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill	4/9/2014	5/25/2014
Trip Drill	4/8/2014	5/29/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	13:00	13....	16"	D	03	P	Directional drilling 16" hole section in rotation and slide mode f/ 1441m to 1478m w/ 800-950 gpm/ 1850-2350 psi/ 30-37 kips/ 35-70 rpm/ 8-17 kft-lbf. Due to Low ROP, make decision to POOH. Slide Interval 1441m - 1443m Rotation Interval 1443m - 1478m ** Drilling with NL / NG. ** B/R and R/D each stand drilled once. ** Pump 20 bbl Lo-Vis + 20 bbl Hi-Vis pill when stand down. ** Take survey before connection. >>> Offline: * M/U Pack-off tools to RT. * M/U Unihead WH to RT. * M/U unihead side outlet valves.
13:00	21:30	8.50	16"	D	04	P	Flow check, well static. POOH directional BHA#5 f/ 1478m to surface. L/D 8" jar. Observe Motor Sleeve stabilizer back off and was Loose (not fully open). B/O the bit (2-8-RO-G-X-1/16-BT-PR). ** 14 3/4" String stabilizer: Partially balled up. ** 15 3/4" Sleeve stabilizer: Partially balled up. ** 9 5/8" Motor bearing gap: 4 mm ** 16" Bit: Partially balled up.
21:30	00:00	2.50	16"	D	24	N	(NPT for HPOGC Company) Open motor sleeve stabilizer (15 3/4") by difficulty. Try to M/U new 15 7/8" Sleeve stabilizer, no success. Decide to change motor.

Daily Drilling Report
SPD19B-03
4/13/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00						Apr 14 , 2014 00:00 - 01:30 (NPT for HPOGC Company) L/D used 9 5/8" motor. P/U new 9 5/8" motor, B/O and L/D Sleeve Protector. M/U new 15 3/4" Sleeve stabilizer. 01:30 - 02:30 RIH directional BHA#6 consists of: New 16" Insert bit (CR14JMRS, 435, SN: 1244888, Nozzles: 3x20 + 1x16) + New 9 5/8" Motor (BH: 1.25 , Sleeve stabilizer: 15 3/4") + 14 3/4" String stabilizer + 9 1/2" Restrictor + 9 1/2" NMDC + 9 1/2" TM - Sub , to 24m. Align motor with MWD. 02:30 - 03:00 (NPT for HPOGC Company) Connect TDS. Shallow test new mud Motor (730 gpm, 850 psi) , OK. 03:00 - 05:00 Cont' RIH rest of BHA consists of: 1 jt 9 1/2" NMDC + 9 1/2" Float sub + XOS + 8 jts 8 1/4" DC + 8" Jar + 6 jts 8 1/4" DC + XOS + 15 jts 5 7/8" HWDP , to 313m. 05:00 - 06:00 RIH directional BHA#6 on 5 7/8" DP f/ 313m to 700m.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,478.00	Polymer Mud	9.30	43	11.0	19.000	7.2
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	92.0	8.0	24,200.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	7.000		0/100		1704.0	1797.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
2245.0	153.0	0.0			13.39637	0.396

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	7.	340	10.0	1,006.0	29.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Navis King	Supply Boat	4/14/2014 01:00		Stdby boat.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bio-Cide	2.0	274.0	103.0
AMYLOSE B	27.0	800.0	363.0
Brine Cor (Corrosion Inhibitor)	1.0	8.0	3.0
Defoam SLC	3.0	220.0	64.0
H2S Scavenger	4.0	120.0	45.0
Bio-XCD	9.0	454.0	134.0
Lime	12.0	338.0	128.0
Caustic Soda	29.0	825.0	236.0
Pac-UL	9.0	230.0	78.0
Pac-R	18.0	360.0	101.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	15.0
Barite Bulk	0.0	0.0	147.0
Cement G	0.0	0.0	168.0
Bentonite Bulk	0.0	0.0	65.0
Bentonite BB	0.0	0.0	21.0
Diesel Fuel	7.0	0.0	122.0
Potable Water	20.0	20.0	110.0
Drill Water	60.0	0.0	413.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
5	16	Varel	VM516DGXU	6004481	PDC	M322	13/13/13/13/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.04	74.38	699,164.40	878.00	1,478.00	600.00	8.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	1,000	1,650.0	2,500.0	10	37	104	194
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
91.20	57.55	77.37	22.63	353	315	237	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,441.00	1,443.00	0.62		850		2,150.0	15	315
1,443.00	1,445.00	0.17	17,000.0	800	40	1,850.0	25	315
1,445.00	1,474.00	6.16	18,000.0	850	40	1,850.0	35	315

Daily Drilling Report
SPD19B-03
4/13/2014

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,474.00	1,478.00	4.87	15,000.0	950	70	2,350.0	37	315

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	139.60	313.30
XOS	8	3.000	0.79	173.70
6 jts 8 1/4" DC	8 1/4	2.810	56.24	172.91
8" DA Drilling Jar (COR2898AD)	8	3.500	5.72	116.67
8 jts 8 1/4" DC	8 1/4	2.810	75.34	110.95
XOS	9 1/2	3.000	1.22	35.61
9 1/2" Float sub with valve	9 1/2	3.000	0.78	34.39
9 1/2" NMDC	9 1/2	4.000	9.44	33.61
9 1/2" TM-Sub (Mwd and gamma probes inside)	9 1/2		2.11	24.17
9 1/2" NMDC	9 1/2	4.000	9.44	22.06
9 1/2" Restrictor Sub (Jet: 36)	9 1/2	4.300	0.92	12.62
14 3/4" String Stabilizer (COR3032AD)	14 3/4	3.000	2.16	11.70
9 5/8" Motor (BH: 1.25 , Sleeve: 15 3/4")	9 5/8		9.14	9.54
16" PDC Bit (SN: 6004481)	16		0.40	0.40

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	1,443.00	14.00	20	120.0	97	114
2	1,443.00	14.00	30	179.0	97	170
2	1,443.00	14.00	40	262.0	97	227
3	1,443.00	14.00	20	117.0	97	114
3	1,443.00	14.00	30	173.0	97	170
3	1,443.00	14.00	40	263.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
1,454.40	248.80	16.80	1,388.39	-174.11	-343.28	384.91	0.09	0.00

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Lower Dariyan		1,183.50	1,132.88
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,176.98
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.82

General Remarks

Comment

>>> Requirements: NIL

>>> Sent: NIL

>>> Received: NIL